

Smart Snippet System — Product Requirements Document (MVP)

Version: 1.0 **Date:** 2026-05-12 **Status:** Draft — Ready for Engineering & Design Review

Audience: Engineering + Design + Product **Related Docs:**

[Smart-Snippet-System-Analysis.md](#)

[Competitive-Landscape-Analysis.md](#)

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Executive Summary .1

Vision

A Customer Intent Intelligence layer for Salla stores — captures *why* visitors don't buy, *what* they're waiting for, and *what* to fix on each product.

MVP Scope

4 Features:

Hesitation Capture Widget (Why) — Exit intent + Product Clarity surveys	.1
Variant & Price Interest Capture (Wait) — Multi-dimensional alerts	.2
Intent Dashboard + Product Doctor Lite (Fix) — Aggregated insights + rule-based recommendations	.3
Widget Analytics (Measure) — Per-widget performance metrics	.4

Target Launch

Beta: Day 60 (closed, 3-5 stores) ●

App Store Submission: Day 90

-

Public Launch: Day 105

-

Target Market (MVP)

Saudi Arabian Salla stores in Beauty, Perfumes, Fashion. 2,000-30,000 monthly visits. 50-500 SKUs.

Pricing

99 / 199 / 349 SAR per month. 14-day free trial, no credit card.

Background & Context .2

Problem Statement

Salla merchants currently know **what** happened in their stores (orders, sales, reviews). They don't know **why** visitors didn't buy on a per-product basis. Existing tools either show display popups (PopupSnap), analyze post-purchase data (SmartInsight, Smart Analytics), or send restock alerts (Salla native). No tool captures pre-purchase intent and turns it into actionable per-product recommendations.

Market Gap

0 competitors in Salla App Store offering Customer Intent capture

-

0 competitors offering Product Doctor / Pre-purchase diagnosis

-

Salla Smart Analytics is post-purchase only (verified)

-

SmartInsight is post-purchase analytics chat (\$184 USD/month) — different positioning

-

Shopify has Simplify Exit Intent Survey (\$4.99) but no Salla integration

-

Why Now

Rising ad costs in Gulf market = merchants need to reduce conversion waste

-

Salla merchant base growing rapidly

-

SmartInsight at 693 SAR proves WTP for intelligence tools

•

6-12 month window before category gets crowded

•

Goals & Non-Goals .3

MVP Goals (3 main)

G1 — Prove Behavioral Hypothesis Verify that Salla store visitors will respond to hesitation surveys at $\geq 3\%$ submission rate.

G2 — Prove Willingness to Pay Validate that merchants will pay 99-349 SAR/month for pre-purchase intent insights.

G3 — Establish Defensible Positioning Build Salla App Store presence in Customer Intent category as first-mover.

Success Targets (90 days post-launch)

50+ stores active (free trial + paid)

•

10+ paid merchants

•

$\geq 3\%$ submission rate on Hesitation widget

•

$\geq 70\%$ 30-day retention

•

3 published case studies

•

Explicit Non-Goals (Out of MVP)

✗ AI-powered recommendations (rule-based only in MVP — AI in Phase 3)

•

✗ Popup builder / general marketing tool functionality

•

✗ Helpdesk / FAQ management

•

✗ Wishlist standalone feature

•

✗ Loyalty / membership programs

•

✗ Post-purchase flows (NPS, reviews, etc.)

•

✗ Multi-store management (single store per account)

•

- ✗ White-label / agency features •
 - ✗ Mobile app (web dashboard only) •
 - ✗ Direct SMS sending (use Salla's existing channels) •
 - ✗ Email marketing automation (basic digest only) •
 - ✗ A/B testing engine •
 - ✗ Conversion attribution / revenue tracking (Phase 2) •
 - ✗ Multi-language beyond Arabic + English •
-

Personas .4

Primary: The Salla Merchant

Name: Sara Al-Otaibi **Age:** 32 **Role:** Founder / Owner of beauty store on Salla **Store stats:**

- 15,000 monthly visits •
- 200 SKUs •
- ~3% conversion rate •
- 25,000 SAR/month ad spend (Meta + TikTok) •

Goals:

- Increase conversion without raising ad budget •
- Understand why visitors don't buy •
- Make better inventory decisions (which colors/sizes to reorder) •
- Reduce dependence on agency/consultant •

Frustrations:

- Salla reports show numbers but not reasons •
- WhatsApp messages are unorganized •
- Doesn't know which products need attention •
- Marketing dashboards are intimidating •

Tech Literacy: Medium. Can install apps, use Salla dashboard, configure basic settings.
Doesn't write code or use complex analytics tools.

Quote: "عندي 200 منتج، ما عندي وقت أحلل كل واحد. خليني أعرف اللي يحتاج اهتمام"

Daily Workflow Today:

- 9:00 AM: Check Salla dashboard, orders count ●
- 11:00 AM: Reply to WhatsApp messages ●
- 2:00 PM: Update product listings ●
- 5:00 PM: Review ad performance ●

Where Smart Snippet Fits: 9:30 AM — Opens Smart Snippet dashboard, reads weekly digest, applies 2 Doctor recommendations.

Secondary: The End Customer

Name: Fatima **Age:** 27 **Behavior:** Browses Salla stores on mobile (85% of sessions), often during evening hours.

Buying Pattern:

- Visits 3-5 stores per shopping session ●
- Adds to cart 40% of the time ●
- Completes checkout 15% of the time ●
- Reasons for not buying: price comparison, sizing uncertainty, waiting for sale ●

Where Smart Snippet Fits: After 60 seconds on a product, scrolling without adding to cart, a non-intrusive question appears: "شو ناقص في هذه الصفحة؟"

She taps "دليل المقاسات" and submits in 5 seconds. The merchant gets the signal.

User Stories .5

Merchant Stories

- M1.** As a Salla merchant, I want to install Smart Snippet System from the Salla App Store in one click so I can start collecting intent data without technical setup.
- M2.** As a Salla merchant, I want a guided 5-minute onboarding that activates my first widget so I can see value within 24 hours.
- M3.** As a Salla merchant, I want to see why visitors abandoned my top 5 products this week so I can prioritize fixes.
- M4.** As a Salla merchant, I want Product Doctor to suggest specific actions (not just data) so I know what to do without an analyst.
- M5.** As a Salla merchant, I want a weekly email summary so I don't need to log in daily.
- M6.** As a Salla merchant, I want to see which variants (colors/sizes) customers want most so I can make reorder decisions.
- M7.** As a Salla merchant, I want to customize hesitation reasons in Arabic and English so they match my store vocabulary.
- M8.** As a Salla merchant, I want my customers to be able to set price alerts so I capture price-sensitive demand.
- M9.** As a Salla merchant, I want a snippet analytics tab to verify widgets are performing ($\geq 3\%$ submission rate).
- M10.** As a Salla merchant, I want a free 14-day trial without credit card so I can test risk-free.

Customer Stories

- C1.** As a Salla store visitor, I want to give quick feedback (one tap) about why I'm not buying so I help the merchant without filling forms.
- C2.** As a visitor interested in a product but not ready to buy, I want to set a price alert so I'm notified when it drops.
- C3.** As a visitor where my size/color isn't available, I want to register interest so I'm notified when restocked.

C4. As a visitor, I want to know my data is protected (PDPL) and I can withdraw consent any time.

C5. As a mobile visitor, I want widgets that don't interrupt my browsing experience.

MVP Scope .6

What's In (Build These 4 Features)

Priority	Modules Combined	Feature	#
P0 (Core)	M4 + M17	Hesitation Capture Widget	1
P0 (Core)	M7 + M8 (variants/price/follow only, not restock)	Variant & Price Interest Capture	2
P0 (Core)	M32 + M34	Intent Dashboard + Product Doctor Lite	3
P0 (Infrastructure)	M35	Widget Analytics	4

What's Out (Defer to Phase 1+)

Why Out	Module
Overlap with Salla Q&A native — needs different positioning	M5 (Ask About Product)
Overlap with "أعلمني عند التوفر" native	M8 Restock-only dimension
Overlap with Salla abandoned cart native	M22 (Cart Rescue)
Phase 1+	Everything else (Modules 1-3, 6, 9-16, 18-21, 23-31, 33, 36)

Feature 1 — Hesitation Capture Widget .7

Description 7.1

A non-intrusive widget that asks visitors why they're not converting, captures structured responses, and aggregates them into merchant insights.

Two trigger modes (configurable per merchant):

Mode A: Exit Intent — Shows when visitor signals intent to leave PDP or Cart

Mode B: Product Clarity — Shows after scroll without add-to-cart on PDP

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-

User Flow 7.2

```
[Customer visits PDP]
  ↓
[Browses 30+ seconds, no add to cart]
  ↓
[Trigger fires]
  ├── Desktop: mouseleave event toward top
  └── Mobile: scroll-up direction + idle 5s
  ↓
[Widget appears as modal/sheet]
  ↓
[Customer taps one option] → Submit
      OR
[Customer types in "Other"] → Submit
      OR
[Customer dismisses] → Track dismissal event
  ↓
[Thank you message: 2 seconds]
  ↓
[Widget closes]
```

Functional Requirements 7.3

FR-1.1 Widget MUST render on Product Detail Page (PDP) and Cart pages only (configurable per merchant).

FR-1.2 Widget MUST support two trigger modes (exit intent / clarity) configurable independently per page type.

FR-1.3 Widget MUST NOT trigger more than **once per session** per page type (deduplication via sessionStorage).

FR-1.4 Widget MUST NOT trigger for customers who have already submitted in last 30 days (cookie or customer-id based).

FR-1.5 Widget MUST display 6 default reason options + 1 free text option, all configurable per merchant.

FR-1.6 Widget MUST submit response to backend within 200ms (optimistic UI: show "thank you" immediately on tap).

FR-1.7 Widget MUST capture metadata: product_id, page_url, session_id, customer_id (if logged in), timestamp, device_type, referrer, UTM params.

FR-1.8 Widget MUST be dismissible via X button and ESC key (desktop).

FR-1.9 Widget MUST NOT block the underlying page (no full-screen overlay on mobile — use bottom sheet instead).

FR-1.10 Widget MUST support Arabic and English with automatic detection from Salla store locale.

FR-1.11 Widget MUST collect PDPL-compliant consent if any PII is requested (in Hesitation widget MVP, no PII = no consent needed unless merchant adds optional contact field).

FR-1.12 Free text input MUST be limited to 500 characters and sanitized server-side.

UX Requirements 7.4

Visual Design:

- Minimal, non-aggressive (NOT popup-styled) •
- Inherits Salla theme colors via Twilight design tokens where possible •
- Rounded corners (8-12px) •
- Subtle drop shadow •
- No bright colors / urgency cues •

Animation:

- Slide-up from bottom on mobile (300ms ease-out) •
- Fade-in on desktop (200ms) •

Slide-down dismiss



Accessibility:

ARIA labels on all interactive elements



Keyboard navigable (Tab through options, Enter to select)



Focus trap when open



Screen reader announcements (`aria-live="polite"` for thank you message)



Color contrast WCAG AA minimum



Responsive:

Desktop: Centered modal max-width 480px



Tablet: Modal with side margins



Mobile: Bottom sheet, full width



Configuration Options (Per Merchant) 7.5

```
interface HesitationWidgetConfig {
  enabled: boolean;
  triggers: {
    exitIntent: {
      enabled: boolean;
      pages: ('pdp' | 'cart')[];
      minTimeOnPage: number; // seconds, default 30
    };
    productClarity: {
      enabled: boolean;
      scrollThreshold: number; // %, default 50
      idleTime: number; // seconds, default 60
    };
  };
  reasons: {
    id: string;
    label_ar: string;
    label_en: string;
    enabled: boolean;
    order: number;
  }[];
  allowFreeText: boolean;
  freeTextPlaceholder_ar: string;
  freeTextPlaceholder_en: string;
  thankYouMessage_ar: string;
  thankYouMessage_en: string;
  showRewardCoupon: boolean; // optional incentive
  couponCode?: string; // if showRewardCoupon
}
```

Default Reasons (Beauty/Perfumes vertical):

1. السعر مرتفع / Price too high
2. الشحن غير واضح / Unclear shipping
3. لست متأكد من المقاس/الحجم / Unsure about size
4. مختار بين منتجات / Comparing products
5. محتاج معلومات أكثر / Need more info
6. فقط أتصفح / Just browsing

Exit Intent — Desktop:

```
// Pseudo-code
document.addEventListener('mouseleave', (e) => {
  if (e.clientY < 20 && !triggered) {
    triggered = true;
    showWidget();
  }
});
```

Exit Intent — Mobile (alternative):

```
// Idle + scroll-up detection
let lastScrollY = window.scrollY;
let idleTimer;

window.addEventListener('scroll', () => {
  const currentY = window.scrollY;
  if (currentY < lastScrollY - 100 && !triggered) {
    triggered = true;
    showWidget();
  }
  lastScrollY = currentY;

  clearTimeout(idleTimer);
  idleTimer = setTimeout(() => {
    if (!triggered && timeOnPage > 30) {
      showWidget(); // backup: idle 30s
    }
  }, 30000);
});
```

Product Clarity (scroll-based):

```
// Fires after scrolling past 50% + 60s on page
const scrollListener = () => {
  if (scrollPercent() > 50 && timeOnPage > 60 && !triggered) {
    triggered = true;
    showWidget();
  }
};
```

Data Captured 7.7

```
interface HesitationSubmission {
  id: string; // UUID
  merchant_id: string;
  widget_id: string;
  trigger_mode: 'exit_intent' | 'product_clarity';
  reason_id?: string; // if predefined
  reason_label?: string; // computed at submission time
  free_text?: string; // sanitized
  product_id: string;
  product_handle: string;
  product_price: number;
  page_url: string;
  page_type: 'pdp' | 'cart';
  customer_id?: string; // if logged in to Salla
  session_id: string;
  device_type: 'desktop' | 'mobile' | 'tablet';
  user_agent: string;
  referrer?: string;
  utm_source?: string;
  utm_medium?: string;
  utm_campaign?: string;
  time_on_page_seconds: number;
  scroll_depth_percent: number;
  created_at: timestamp;
  ip_country?: string; // for analytics, not stored as PII
}
```

Edge Cases 7.8

Behavior	Edge Case
Don't show widget. Track skip in analytics.	Customer already submitted today
Don't render widget.	Page has no product data (404, custom)
Throttle to avoid CPU spike.	Customer scrolling fast (>10 events/s)
Each PDP can trigger independently (max 3 per session).	Multiple PDPs in single session
Widget doesn't render. Graceful degradation.	Customer with JavaScript disabled
Widget uses Shadow DOM to isolate styling.	Salla theme overrides body styles
Submission queues to localStorage, retries on next page load.	Slow network (3G)
Show "تم الاستلام" anyway (optimistic), retry in background up to 3 times.	Submission fails (500 error)

Mobile Considerations 7.9

- Bottom sheet instead of modal
- Touch-friendly buttons (44px min height)
- Keyboard-aware for free text (scroll to keep input visible)
- Swipe-down to dismiss
- No exit intent via mouseleave (use scroll-up + idle instead)

Accessibility 7.10

- All buttons have `role="button"` and accessible labels
- Modal has `role="dialog"` + `aria-modal="true"`
- Focus moves to first option on open

Focus returns to trigger element on close

Color contrast minimum 4.5:1 for text

-
-

Feature 2 — Variant & Price Interest Capture .8

Description 8.1

Captures latent interest from visitors who don't buy due to price, missing variant (color/size), or wanting to follow updates. Aggregates into Multi-Dimensional Interest Graph.

User Flow 8.2

```
[Customer on PDP]
  ↓
[Sees "نبهني" button next to Add to Cart]
  ↓
[Taps button]
  ↓
[Modal shows interest options]
  ↓
[Customer selects: price drop / variant / follow]
  ↓
[Enters phone number (Saudi format)]
  ↓
[PDPL consent checkbox (unchecked by default)]
  ↓
[Submit] → Confirmation
  ↓
[Customer receives WhatsApp/SMS when trigger fires]
```

Functional Requirements 8.3

FR-2.1 "نبهني" button MUST render on PDP, positioned near Add to Cart button.

FR-2.2 System MUST support 4 interest types: target_price, variant_color, variant_size, follow_product.

FR-2.3 System MUST support follow_category and follow_brand (toggleable, off by default in MVP).

FR-2.4 Customer MUST enter Saudi/Gulf phone format (validation: +966, +971, +965, +973, +968, +974).

FR-2.5 Customer MUST explicitly check PDPL consent checkbox (unchecked default).

FR-2.6 System MUST NOT send notifications until consent timestamp is stored.

FR-2.7 When target_price condition met (via `product.price.updated` webhook), system MUST queue notification to customer.

FR-2.8 When variant restock condition met (via `product.updated` webhook), system MUST queue notification.

FR-2.9 Notifications MUST be sent via WhatsApp Business API (primary) with SMS fallback.

FR-2.10 Each customer MUST be able to view and delete their interest signals via a public consent management URL.

FR-2.11 Variant detection MUST work with Salla product options (color, size, etc.) — handle stores with custom options.

FR-2.12 System MUST deduplicate interest signals (same customer + same product + same condition = 1 record).

UX Requirements 8.4

Button Style:

Secondary button (not primary — doesn't compete with Add to Cart)

Icon:  or bell SVG

Text: "نبهني" / "Notify me"

Position: below or beside Add to Cart

Modal Style:

Slide-up bottom sheet on mobile

Centered modal on desktop (max-width 420px)

Clear sections: choose interest → phone → consent

Trust Signals in Modal:

"بياناتك محمية وفق نظام PDPL" 

• "👉 يمكنك إلغاء التنبيهات في أي وقت"

Configuration Options 8.5

```
interface InterestWidgetConfig {
    enabled: boolean;
    buttonPosition: 'next_to_add_to_cart' | 'below_add_to_cart' | 'sticky_bottom';
    buttonLabel_ar: string;
    buttonLabel_en: string;
    enabledInterestTypes: {
        targetPrice: boolean;
        variantColor: boolean;
        variantSize: boolean;
        followProduct: boolean;
        followCategory: boolean; // off by default
        followBrand: boolean; // off by default
    };
    notificationChannels: {
        whatsapp: boolean; // requires merchant WA Business setup
        sms: boolean;
        email: boolean;
    };
    defaultMessage_ar: string;
    defaultMessage_en: string;
    consentText_ar: string;
    consentText_en: string;
    privacyPolicyUrl: string;
}
```

```
interface InterestSignal {
  id: string;
  merchant_id: string;
  customer_phone: string; // hashed in storage, last 4 visible
  customer_email?: string;
  customer_salla_id?: string;
  product_id: string;
  product_variant_id?: string;
  interest_type: 'target_price' | 'variant_color' | 'variant_size' | 'follow_p
  target_value?: {
    price?: number;
    color?: string;
    size?: string;
    category_id?: string;
    brand_id?: string;
  };
  status: 'active' | 'triggered' | 'cancelled' | 'expired';
  consent_given_at: timestamp;
  consent_ip: string;
  consent_text_version: string;
  notification_channel: 'whatsapp' | 'sms' | 'email';
  notification_sent_at?: timestamp;
  notification_clicked_at?: timestamp;
  cancelled_at?: timestamp;
  expires_at: timestamp; // default: 90 days
  created_at: timestamp;
}
```

Edge Cases 8.7

Behavior	Edge Case
Update existing record, don't create duplicate	Phone already registered for this product
Reject with clear error message	Customer enters non-Saudi/Gulf phone
Inline error, don't submit	Phone validation: invalid format
Submit button disabled	Consent unchecked
Show warning: "السعر الحالي أقل من الذي حددته"	Target price $\leq$ current price
Notify customer that variant was discontinued	Variant doesn't exist anymore
Cascade delete all their signals for this merchant	Customer hits delete endpoint
Skip notification	Webhook arrives after customer cancelled
Fallback to SMS, then email	Notification fails (WA blocked)

Integration with Salla Webhooks 8.8

```
// Pseudo-code for backend webhook handler
on('product.price.updated', async (event) => {
  const { product_id, new_price, old_price } = event.data;

  if (new_price ≥ old_price) return; // only on price drops

  const matchingSignals = await db.interestSignals.find({
    product_id,
    interest_type: 'target_price',
    status: 'active',
    'target_value.price': { $gte: new_price },
  });

  for (const signal of matchingSignals) {
    await notificationQueue.enqueue({
      signal_id: signal.id,
      channel: signal.notification_channel,
      template: 'price_drop',
      params: { product_id, new_price, old_price },
    });
  }
});

on('product.updated', async (event) => {
  // Check variant availability changes
  // Match against variant_color and variant_size signals
});

on('product.quantity.low', async (event) => {
  // Note: restock not part of MVP (Salla native handles)
  // But low quantity could trigger urgency signals later
});
```

Mobile Considerations 8.9

Phone input: numeric keyboard

Country code dropdown shows flag

Submit button: full width on mobile

-
-
-

Accessibility 8.10

- Form labels associated with inputs (htmlFor) •
- Error messages announced via `aria-live` •
- Consent checkbox: clear label, large tap target •

Feature 3 — Intent Dashboard + Product Doctor .9

Description 9.1

Merchant-facing dashboard that aggregates hesitation submissions and interest signals into actionable insights, with a rule-based Product Doctor that recommends specific fixes per product.

Dashboard Sections 9.2

Section A: Top Stats (Hero)

This Week:

- Hesitation Submissions: 142 (↑18% vs last week)
- Interest Signals: 87 (↑22%)
- Active Widgets: 4 / 5
- Submission Rate: 3.1% 

Section B: Top 3 Hesitation Reasons Bar chart with %, raw count. Filter: This Week / Last 30 Days / All Time.

Section C: Products Needing Attention Table sorted by total intent signals (hesitation + interest). Click row → product detail view.

Product	Issues	Severity
Vitamin C Serum	15	 High
Black Maxi Dress	12	 Medium
Oud Royal 100ml	9	 Medium
Niacinamide Serum	7	 Low

Section D: Variant Heatmap Shows most requested variants (colors/sizes) across all products.

Section E: Doctor Recommendations Card list of active recommendations. Each card has Apply button.

Doctor Rules (5 Rules MVP) 9.3

```
interface DoctorRule {
  id: string;
  name: string;
  condition: (signals: ProductSignals) ⇒ boolean;
  recommendation: {
    title_ar: string;
    title_en: string;
    explanation_ar: string;
    explanation_en: string;
    actionType: 'create_snippet' | 'reorder_inventory' | 'create_bundle' | 'ad
    actionPayload: any;
  };
  severity: 'low' | 'medium' | 'high';
}

const MVP_RULES: DoctorRule[] = [
  {
    id: 'rule_high_price_complaints',
    name: 'High Price Complaints',
    condition: (s) ⇒ s.hesitationByReason['price_high'] ≥ 5,
    recommendation: {
      title_ar: 'عميل X السعر يبدو مرتفعًا لـ',
      title_en: 'Price seems high to X customers',
      explanation_ar: 'بدلاً من خصم، أو اعرض ضمان السعر bundle فكر في',
      explanation_en: 'Consider a bundle instead of discount, or offer price g
      actionType: 'create_bundle',
      actionPayload: { product_id: '$product_id', suggested_bundle_value: 0.15
    },
    severity: 'high',
  },
  {
    id: 'rule_size_confusion',
    name: 'Size Confusion',
    condition: (s) ⇒ s.hesitationByReason['size_unclear'] ≥ 5,
    recommendation: {
      title_ar: 'عميل غير متأكد من المقاس X',
      title_en: 'X customers unsure about size',
      explanation_ar: 'أو دليل المقاسات Size Help snippet أضف',
      explanation_en: 'Add a Size Help snippet or size chart',
      actionType: 'create_snippet',
      actionPayload: { snippet_template: 'size_help' },
    },
    severity: 'high',
  },
  {
    id: 'rule_info_missing',
    name: 'Information Missing',
```



```

condition: (s) => s.hesitationByReason['need_more_info'] ≥ 5 || s.clarity
recommendation: {
  title_ar: 'صفحة المنتج تحتاج معلومات أكثر',
  title_en: 'Product page needs more info',
  explanation_ar: 'أضف FAQ أو فيديو شرح',
  explanation_en: 'Add FAQ or explanation video',
  actionType: 'add_content',
  actionPayload: { content_type: 'faq_or_video' },
},
severity: 'medium',
},
{
  id: 'rule_price_demand',
  name: 'Price Drop Demand',
  condition: (s) => s.interestByType['target_price'] ≥ 5,
  recommendation: {
    title_ar: 'X عميل ينتظر نزول السعر',
    title_en: 'X customers waiting for price drop',
    explanation_ar: 'بدلاً من خصم لتحويلهم bundle جرب',
    explanation_en: 'Try a bundle instead of discount to convert them',
    actionType: 'create_bundle',
    actionPayload: { audience: 'price_watchers' },
  },
  severity: 'medium',
},
{
  id: 'rule_variant_demand',
  name: 'Variant Demand',
  condition: (s) => s.interestByType['variant_color'] + s.interestByType['va
recommendation: {
  title_ar: 'X غير متوفر variant عميل يطلب',
  title_en: 'X customers requesting unavailable variant',
  explanation_ar: 'للمخزون reorder فكر في',
  explanation_en: 'Consider reordering inventory',
  actionType: 'reorder_inventory',
  actionPayload: { suggested_quantity: '$signal_count * 2' },
},
  severity: 'high',
},
];

```

Recommendations Engine 9.4

Pipeline:

- Daily Cron Job** runs at 4 AM (Saudi time) .1
- For each active merchant, computes per-product signals from last 7 days .2

- Evaluates each rule against each product's signals .3
- Generates new Recommendation records (status: `pending`) .4
- Deduplicates against existing pending/applied recommendations (don't re-recommend the same thing within 7 days) .5
- Sends weekly digest email Monday 9 AM .6

Recommendation Lifecycle:

- `pending` → merchant sees in dashboard •
- `applied` → merchant clicked Apply •
- `dismissed` → merchant clicked Dismiss •
- `expired` → 14 days without action •

Apply Actions (MVP):

- `create_snippet` : Opens snippet creation wizard with template pre-filled •
- `reorder_inventory` : Exports CSV with suggested variant + quantity •
- `create_bundle` : Opens Salla bundle creation in new tab (link to Salla admin) •
- `add_content` : Shows checklist of suggested content additions •

Weekly Email Digest 9.5

Send Time: Monday 9 AM Saudi time **Subject:** "ملخص أسبوع متجرك — 3 توصيات جاهزة 📊"

Content Template:

[Merchant Name] السلام عليكم

:لمتجرك هذا الأسبوع intent signals هذا ملخص

📧 Submissions: 142 (↑18%)

🔔 Interest Signals: 87 (↑22%)

:أهم 3 أسباب التردد

1. السعر مرتفع - 54 رد
2. غير متأكد من المقاس - 32
3. معلومات ناقصة - 28

🩺 جاهزة Doctor توصيات

1. Vitamin C Serum: أضف فيديو شرح
2. Black Dress L: 30 طلب - reorder priority
3. Oud Royal 100ml: 18 جَرَب - سعر - bundle

[افتح Dashboard للتفاصيل →]

Functional Requirements 9.6

FR-3.1 Dashboard MUST load in <2 seconds on 3G connection (target <500ms on 4G).

FR-3.2 All metrics MUST refresh every 5 minutes (or on-demand via refresh button).

FR-3.3 Dashboard MUST support date range filtering: 7d, 30d, 90d, all-time.

FR-3.4 Doctor Recommendations MUST NOT show same recommendation for same product more than once every 7 days (unless threshold doubles).

FR-3.5 Apply action MUST be one-click where possible.

FR-3.6 All numbers MUST be clickable (drill down to source data).

FR-3.7 Weekly digest MUST send to merchant's primary email (from Salla store profile).

FR-3.8 Dashboard MUST be mobile-responsive (merchant might check on phone).

UX Requirements 9.7

Information Hierarchy:

- | | |
|---|----|
| What's urgent (red flags) — top of page | .1 |
| What's actionable (Doctor recommendations) — next | .2 |
| What's interesting (trends, charts) — bottom | .3 |

Visual Design:

- Clean, data-dense (NOT marketing-styled)
- Use color sparingly: red/yellow/green for severity
- Bold numbers, lighter labels
- Action buttons clearly visible

Empty States:

- Day 1: "أول وستظهر بياناتك خلال 24 ساعة widget مرحبًا — ابني 🙌"
- Less than 10 submissions: "🇸🇦 insights جمعنا 7 ردود — نحتاج +10 لإنتاج"
- No recommendations: "كل المنتجات بحالة جيدة هذا الأسبوع ✅"

Edge Cases 9.8

Behavior	Edge Case
Show empty state with onboarding tips	Merchant has 0 submissions
Don't show in Doctor (insufficient data)	Merchant has <10 submissions for product
Suppress that rule for that product for 30 days	Merchant dismisses same rec 3 times
Show both, sorted by severity	Two rules trigger for same product
Pagination (50 per page), lazy load	Merchant's store has 5,000+ products

Feature 4 — Widget Analytics .10

Description 10.1

Per-widget performance metrics. Lightweight in MVP — no funnel analysis, no A/B testing.

Functional Requirements 10.2

FR-4.1 System MUST track these events per widget:

`widget_viewed` (impression)

•

`widget_interacted` (any click/tap)

•

`widget_submitted` (successful submission)

•

`widget_dismissed` (closed without submitting)

•

FR-4.2 Dashboard MUST display per widget:

Views (count)

•

Submissions (count)

•

Submission rate (= submissions / views)

•

Top pages by submissions (top 5)

•

7-day trend chart (sparkline)

•

FR-4.3 Submission rate MUST display visual indicator:

≥3%:  green

•

1.5-3%:  yellow

•

<1.5%:  red with hint "اعد فكرة ال widget"

•

FR-4.4 Analytics MUST be queryable by date range (7d, 30d, all).

Data Model 10.3

```
interface WidgetEvent {
  id: string;
  merchant_id: string;
  widget_id: string;
  event_type: 'viewed' | 'interacted' | 'submitted' | 'dismissed';
  page_url: string;
  page_type: 'pdp' | 'cart' | 'other';
  product_id?: string;
  session_id: string;
  device_type: 'desktop' | 'mobile' | 'tablet';
  created_at: timestamp;
}

// Aggregated views (for performance)
interface WidgetMetricsDaily {
  merchant_id: string;
  widget_id: string;
  date: date;
  views: number;
  interactions: number;
  submissions: number;
  dismissals: number;
}
```

UX Requirements 10.4

Per-widget card layout:

 Exit Survey Widget

Views:	4,532
Submissions:	142
Rate:	3.1% 

 Last 7 days

Top pages:

/products/vitamin-c	24
/products/black-dress	18

[View Details]

Cross-Cutting Requirements .11

Internationalization (i18n) 11.1

- Default languages: Arabic (ar) + English (en)
- Detect from Salla store locale via API
- Allow merchant override in settings
- All UI strings externalized to i18n files
- Date/number formatting per locale
- RTL support for Arabic (use `dir="rtl"` and CSS logical properties)

PDPL Compliance (MANDATORY) 11.2

- Explicit opt-in consent for any PII collection (phone, email)
- Consent checkbox unchecked by default
- Documented consent: timestamp + IP + consent text version + locale
- Privacy Policy URL provided by merchant (we generate template)
- Data Subject Rights endpoint: `/customer/consent/manage?token= ...`
- View all my data
- Delete all my data
- Withdraw consent
- Data retention: 90 days for unconsented data, indefinite (until withdrawal) for consented
- No selling/sharing data with third parties
- Server location: must be Saudi Arabia or compliant region

Performance Budget 11.3

- Snippet JS bundle: ≤ 30 KB gzipped
- Snippet load: ≤ 200 ms on 4G
- Page load impact: ≤ 100 ms (TTI delta)
- API response times:

GET /dashboard: ≤500ms p95	○
POST /widget/submit: ≤200ms p95	○
POST /interest/track: ≤200ms p95	○
Webhook processing: ≤2s p95	●

Mobile Responsiveness 11.4

Mobile-first design	●
Test on iPhone SE (smallest target), iPhone 14, iPhone 14 Pro Max, Galaxy S22, iPad	●
All widgets: touch-friendly (44px tap targets minimum)	●
Dashboard: responsive table → cards on mobile	●

Accessibility (WCAG 2.1 AA) 11.5

Color contrast 4.5:1 minimum for text	●
Keyboard navigation for all interactive elements	●
Focus visible on all focusable elements	●
Screen reader announcements for dynamic content	●
Alt text on all images	●
Form labels associated with inputs	●

Security 11.6

HTTPS everywhere	●
CSP headers (no inline scripts in production)	●
API auth: OAuth 2.0 with Salla + JWT for our backend	●
Rate limiting: 100 req/min per merchant, 10 submissions/min per session	●
Input sanitization on all user-submitted strings (server-side)	●
SQL injection prevention (parameterized queries / ORM)	●
XSS prevention (escape all rendered text)	●
CSRF tokens on state-changing endpoints	●
Sensitive data (phone numbers) hashed at rest, partial display	●
Audit log for all admin actions	●

Browser Support 11.7

Chrome: latest 2 versions	•
Safari: latest 2 versions (iOS Safari critical)	•
Firefox: latest 2 versions	•
Edge: latest 2 versions	•
IE 11: NOT supported	•

High-Level Diagram 12.1



Tech Stack Recommendation 12.2

Frontend (Storefront Widget):

- Vanilla TypeScript or Preact (small footprint) •
- No heavy framework — under 30 KB gzipped •
- Shadow DOM for style isolation •
- Inline SVG for icons •

Frontend (Merchant Dashboard):

- Next.js (SSR + good DX) •
- TypeScript •
- Tailwind CSS (compatible with Salla design tokens) •
- React Query for data fetching •
- shadcn/ui or similar for components •

Backend:

- Node.js (TypeScript) — easy hire pool in Saudi •
- Framework: NestJS or Fastify •
- ORM: Prisma or Drizzle •
- Queue: BullMQ + Redis •
- Cron: node-cron or similar •

Database:

- PostgreSQL 16+ •
- Read replica for analytics queries (optional in MVP) •

Infrastructure:

- Hosting: AWS Riyadh region (PDPL compliance) •
- CDN: CloudFront or Bunny •
- Monitoring: Sentry + Datadog/Grafana •
- Logs: CloudWatch or Loki •

Third-party:

WhatsApp Business API: Meta direct or Wati/Twilio

•

Email: SendGrid or Postmark

•

SMS: Unifonic (Saudi-focused) or Twilio

•

Deployment Strategy 12.3

CI/CD: GitHub Actions

•

Environments: dev / staging / production

•

Database migrations: Prisma Migrate

•

Feature flags: Unleash or simple env vars

•

Zero-downtime deploys: blue-green or rolling

•

Salla Integration Specifications .13

App Registration 13.1

Register as Salla App via Salla Partners Portal:

App name: Smart Snippet System

•

Type: Public (App Store)

•

Redirect URLs: <https://app.smartsnippet.sa/auth/salla/callback>

•

OAuth Scopes Required 13.2

Justification	Purpose	Scope
Phase 2 — request now for future use	Future: post-purchase context	<code>read_orders</code>
Required for Feature 2	Match interest signals to customers	<code>read_customers</code>
Required for all features	Display products in dashboard	<code>read_products</code>
Required	Store merchant config	<code>write_app_settings</code>

Verify exact scope names in Salla docs.

Webhooks Subscribed (MVP) 13.3

Handler	Used For	Webhook Event
<code>webhooks/price-updated.ts</code>	Trigger target_price notifications	<code>product.price.updated</code>
<code>webhooks/product-updated.ts</code>	Detect variant restock	<code>product.updated</code>
<code>webhooks/quantity-low.ts</code> (log only in MVP)	Future use	<code>product.quantity.low</code>
<code>webhooks/customer-created.ts</code>	Match anonymous signals to customers	<code>customer.created</code>
<code>webhooks/app-installed.ts</code>	Onboarding trigger	<code>app.installed</code>
<code>webhooks/app-uninstalled.ts</code>	Cleanup + offboard email	<code>app.uninstalled</code>

Twilight Theme Hooks Used (Confirmed) 13.4

Used For	Hook
Inject main snippet bundle + Exit Intent listener	<code>body:end</code>
Inject CSS for Shadow DOM styles	<code>head:end</code>
Place Interest Capture button + Hesitation widget anchor on PDP	<code>product:single.description.end</code>
Place Hesitation widget on Cart page	<code>cart:items.end</code>

Salla API Endpoints Used (Verify exact paths in Salla Merchant API 13.5 docs)

Purpose	Endpoint
Fetch product details for dashboard	<code>GET /products/:id</code>
Bulk product fetch	<code>GET /products?filter[ids]=...</code>
Match signals to customers	<code>GET /customers/:id</code>
Get store info, locale	<code>GET /store</code>

App Snippet Configuration 13.6

Register App Snippets in Salla Partners Portal:

- Snippet 1: `smart_snippet_main` — `body:end`, all pages •
 - Snippet 2: `smart_snippet_pdp` — `product:single.description.end` •
 - Snippet 3: `smart_snippet_cart` — `cart:items.end` •
-

Schema Diagrams (PostgreSQL) 14.1

```
-- Merchants (Salla stores using our app)
CREATE TABLE merchants (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  salla_store_id VARCHAR(255) UNIQUE NOT NULL,
  store_name VARCHAR(255),
  store_url VARCHAR(255),
  store_locale VARCHAR(10) DEFAULT 'ar',
  store_currency VARCHAR(10) DEFAULT 'SAR',
  store_timezone VARCHAR(50) DEFAULT 'Asia/Riyadh',
  email VARCHAR(255),
  phone VARCHAR(50),
  plan VARCHAR(20) DEFAULT 'trial', -- trial / starter / growth / pro
  trial_ends_at TIMESTAMP,
  subscription_status VARCHAR(20) DEFAULT 'trial', -- trial / active / past_due
  salla_access_token TEXT, -- encrypted
  salla_refresh_token TEXT, -- encrypted
  installed_at TIMESTAMP NOT NULL DEFAULT NOW(),
  uninstalled_at TIMESTAMP,
  created_at TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMP NOT NULL DEFAULT NOW()
);

-- Widgets (one merchant can have multiple widgets)
CREATE TABLE widgets (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  merchant_id UUID NOT NULL REFERENCES merchants(id) ON DELETE CASCADE,
  type VARCHAR(50) NOT NULL, -- 'hesitation' / 'interest' / 'clarity'
  name VARCHAR(255),
  enabled BOOLEAN DEFAULT TRUE,
  config JSONB NOT NULL, -- WidgetConfig (see TypeScript types)
  created_at TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMP NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_widgets_merchant ON widgets(merchant_id);
CREATE INDEX idx_widgets_enabled ON widgets(enabled) WHERE enabled = TRUE;

-- Hesitation Submissions
CREATE TABLE hesitation_submissions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  merchant_id UUID NOT NULL REFERENCES merchants(id) ON DELETE CASCADE,
  widget_id UUID NOT NULL REFERENCES widgets(id) ON DELETE CASCADE,
  trigger_mode VARCHAR(50) NOT NULL, -- 'exit_intent' / 'product_clarity'
```

```

reason_id VARCHAR(100),
reason_label_ar VARCHAR(255),
reason_label_en VARCHAR(255),
free_text TEXT,
product_id VARCHAR(255),
product_handle VARCHAR(255),
product_price DECIMAL(10,2),
page_url TEXT,
page_type VARCHAR(20),
customer_id VARCHAR(255), -- Salla customer ID if logged in
session_id VARCHAR(255),
device_type VARCHAR(20),
user_agent TEXT,
referrer TEXT,
utm_source VARCHAR(255),
utm_medium VARCHAR(255),
utm_campaign VARCHAR(255),
time_on_page_seconds INT,
scroll_depth_percent INT,
ip_country VARCHAR(10),
created_at TIMESTAMP NOT NULL DEFAULT NOW()
);

```

```

CREATE INDEX idx_hesitation_merchant_created ON hesitation_submissions(merchant_id, created_at);
CREATE INDEX idx_hesitation_product ON hesitation_submissions(merchant_id, product_id);
CREATE INDEX idx_hesitation_reason ON hesitation_submissions(merchant_id, reason_id);

```

-- Interest Signals

```

CREATE TABLE interest_signals (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  merchant_id UUID NOT NULL REFERENCES merchants(id) ON DELETE CASCADE,
  customer_phone_hash VARCHAR(255), -- SHA256
  customer_phone_last4 VARCHAR(4),
  customer_email VARCHAR(255),
  customer_salla_id VARCHAR(255),
  product_id VARCHAR(255) NOT NULL,
  product_variant_id VARCHAR(255),
  interest_type VARCHAR(50) NOT NULL, -- 'target_price' / 'variant_color' / etc
  target_value JSONB,
  status VARCHAR(20) DEFAULT 'active', -- 'active' / 'triggered' / 'cancelled'
  consent_given_at TIMESTAMP NOT NULL,
  consent_ip VARCHAR(45),
  consent_text_version VARCHAR(20),
  notification_channel VARCHAR(20) DEFAULT 'whatsapp',
  notification_sent_at TIMESTAMP,
  notification_clicked_at TIMESTAMP,
  cancelled_at TIMESTAMP,
  expires_at TIMESTAMP NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMP NOT NULL DEFAULT NOW(),
  UNIQUE(merchant_id, customer_phone_hash, product_id, interest_type, target_value)
);

```



```

);

CREATE INDEX idx_interest_merchant ON interest_signals(merchant_id);
CREATE INDEX idx_interest_product ON interest_signals(merchant_id, product_id)
CREATE INDEX idx_interest_status ON interest_signals(status) WHERE status = 'a

-- Doctor Recommendations
CREATE TABLE recommendations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  merchant_id UUID NOT NULL REFERENCES merchants(id) ON DELETE CASCADE,
  rule_id VARCHAR(100) NOT NULL,
  product_id VARCHAR(255),
  severity VARCHAR(20) NOT NULL, -- 'low' / 'medium' / 'high'
  title_ar VARCHAR(500),
  title_en VARCHAR(500),
  explanation_ar TEXT,
  explanation_en TEXT,
  action_type VARCHAR(50),
  action_payload JSONB,
  signal_count INT,
  status VARCHAR(20) DEFAULT 'pending', -- 'pending' / 'applied' / 'dismissed'
  applied_at TIMESTAMP,
  dismissed_at TIMESTAMP,
  expires_at TIMESTAMP NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_recommendations_merchant_status ON recommendations(merchant_id, status)
CREATE INDEX idx_recommendations_product ON recommendations(merchant_id, product_id)

-- Widget Events (analytics)
CREATE TABLE widget_events (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  merchant_id UUID NOT NULL,
  widget_id UUID NOT NULL,
  event_type VARCHAR(50) NOT NULL,
  page_url TEXT,
  page_type VARCHAR(20),
  product_id VARCHAR(255),
  session_id VARCHAR(255),
  device_type VARCHAR(20),
  created_at TIMESTAMP NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_events_merchant_widget_date ON widget_events(merchant_id, widget_id, created_at)

-- Aggregated daily metrics (rebuilt nightly)
CREATE TABLE widget_metrics_daily (
  merchant_id UUID NOT NULL,
  widget_id UUID NOT NULL,
  date DATE NOT NULL,

```

```

views INT DEFAULT 0,
interactions INT DEFAULT 0,
submissions INT DEFAULT 0,
dismissals INT DEFAULT 0,
PRIMARY KEY (merchant_id, widget_id, date)
);

-- Consent Records (PDPL)
CREATE TABLE consent_records (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  merchant_id UUID NOT NULL,
  customer_phone_hash VARCHAR(255),
  customer_email VARCHAR(255),
  consent_text TEXT NOT NULL,
  consent_text_version VARCHAR(20),
  consent_ip VARCHAR(45),
  consent_user_agent TEXT,
  consent_locale VARCHAR(10),
  given_at TIMESTAMP NOT NULL DEFAULT NOW(),
  withdrawn_at TIMESTAMP
);

```

Data Retention Policy 14.2

- Hesitation submissions (no PII): retained indefinitely (anonymized after 12 months) •
- Interest signals with consent: until customer withdraws or expires (90 days default) •
- Interest signals without consent: cannot exist (consent required at creation) •
- Widget events: 12 months raw, then aggregated only •
- Consent records: 7 years (legal compliance) •
- Merchant data: deleted 90 days after uninstall •

API Specifications .15

Public API (Storefront — called from snippet) 15.1

Base URL: <https://api.smartsnippet.sa/v1/public>

POST /submissions

Submit a hesitation response.

```
POST /v1/public/submissions
Content-Type: application/json
X-Merchant-Token: <store-specific public token>

{
  "widget_id": "uuid",
  "trigger_mode": "exit_intent",
  "reason_id": "price_high",
  "free_text": null,
  "product": {
    "id": "12345",
    "handle": "vitamin-c-serum",
    "price": 199.00
  },
  "page": {
    "url": "https://store.salla.sa/products/vitamin-c-serum",
    "type": "pdp"
  },
  "session": {
    "id": "anon-session-uuid",
    "device_type": "mobile",
    "time_on_page_seconds": 45,
    "scroll_depth_percent": 67
  },
  "utm": { "source": "instagram", "campaign": "spring2026" },
  "customer_id": null
}

→ 200 OK
{ "submitted": true, "thank_you_message": "شكراً لمساعدتك" }

→ 429 Too Many Requests (rate limit)
→ 400 Bad Request (invalid payload)
```

POST /interest

Register an interest signal (requires consent).

```
POST /v1/public/interest
{
  "product_id": "12345",
  "product_variant_id": "67890",
  "interest_type": "target_price",
  "target_value": { "price": 150.00 },
  "phone": "+966501234567",
  "consent": {
    "given": true,
    "text_version": "v1.0-ar",
    "locale": "ar"
  },
  "notification_channel": "whatsapp"
}

→ 201 Created
{ "interest_id": "uuid", "status": "active" }
```

→ 422 Unprocessable Entity (consent missing or invalid phone)

POST /events

Track widget events (impressions, dismissals).

```
POST /v1/public/events
{
  "widget_id": "uuid",
  "event_type": "viewed",
  "page_url": "...",
  "session_id": "..."
}

→ 202 Accepted (fire and forget)
```

GET /consent/manage

Customer-facing consent management page (HTML).

```
GET /v1/public/consent/manage?token=<jwt>

→ 200 OK (renders HTML page with: view signals, delete signals, withdraw conse
```

Merchant API (Dashboard — authenticated) 15.2

Base URL: <https://api.smartsnippet.sa/v1/merchant> **Auth:** Bearer JWT (issued after Salla OAuth)

GET /me

Get current merchant profile.

GET /dashboard/insights?period=7d

Returns hero stats, top reasons, products needing attention, variant heatmap.

```
GET /v1/merchant/dashboard/insights?period=7d
```

```
→ 200 OK
```

```
{
  "period": "7d",
  "stats": {
    "submissions_total": 142,
    "submissions_delta_percent": 18,
    "interest_signals_total": 87,
    "interest_signals_delta_percent": 22,
    "active_widgets": 4,
    "total_widgets": 5,
    "submission_rate": 0.031
  },
  "top_reasons": [
    { "reason_id": "price_high", "label_ar": "السعر مرتفع", "count": 54 },
    { "reason_id": "size_unclear", "label_ar": "غير متأكد من المقاس", "count": 12 }
  ],
  "products_needing_attention": [
    {
      "product_id": "12345",
      "product_name": "Vitamin C Serum",
      "issue_count": 15,
      "severity": "high"
    }
  ],
  "variant_heatmap": [
    { "color": "Black", "size": "L", "requests": 30 }
  ]
}
```

GET /widgets

List all merchant widgets.

POST /widgets

Create a new widget.

PATCH /widgets/:id

Update widget config.

GET /widgets/:id/analytics?period=7d

Per-widget analytics.

GET /recommendations?status=pending

List Doctor recommendations.

POST /recommendations/:id/apply

Mark recommendation as applied + optionally trigger action.

POST /recommendations/:id/dismiss

GET /submissions?product_id=...&period=30d

List raw submissions (with filters).

GET /interest-signals?status=active

POST /billing/subscribe

Initiate paid plan subscription (links to Salla billing).

Webhook Handlers (Internal) 15.3

Base URL: <https://api.smartsnippet.sa/v1/webhooks/salla>

POST /webhooks/salla/product-price-updated

POST /webhooks/salla/product-updated

POST /webhooks/salla/customer-created

POST /webhooks/salla/app-installed

POST /webhooks/salla/app-uninstalled

All Salla webhooks verified via HMAC SHA256 signature.

UX / UI Requirements .16

Design Principles 16.1

Minimal friction — Don't interrupt the shopping flow	.1
Clear information hierarchy — Most important info first	.2
Action-oriented — Every screen has a clear next step	.3
Data-confident — Don't hide numbers, show them clearly	.4
Salla-native feeling — Inherit theme tokens where possible	.5
Honest empty states — Tell user what's missing and how to fix	.6
No dark patterns — No fake urgency, no hidden costs	.7

Customer-Facing Widget Design 16.2

Color Palette (default — overridable per merchant):

Primary: Inherits Salla theme primary color	•
Background: White / Dark gray (auto-detect)	•
Text: #1A1A1A / #FFFFFF	•
Borders: rgba(0,0,0,0.08)	•
Success: #10B981	•
Error: #EF4444	•

Typography:

Arabic: "IBM Plex Sans Arabic" / fallback to system

English: "Inter" / fallback to system

Sizes: 14px body, 16px buttons, 18px titles

Spacing:

Base unit: 4px

Modal padding: 24px desktop, 16px mobile

Button padding: 12px 24px

Components:

Buttons: rounded-lg (8px), no shadows

Inputs: rounded-md (6px), border 1px

Radio buttons: large circular targets (24px)

Merchant Dashboard Design 16.3

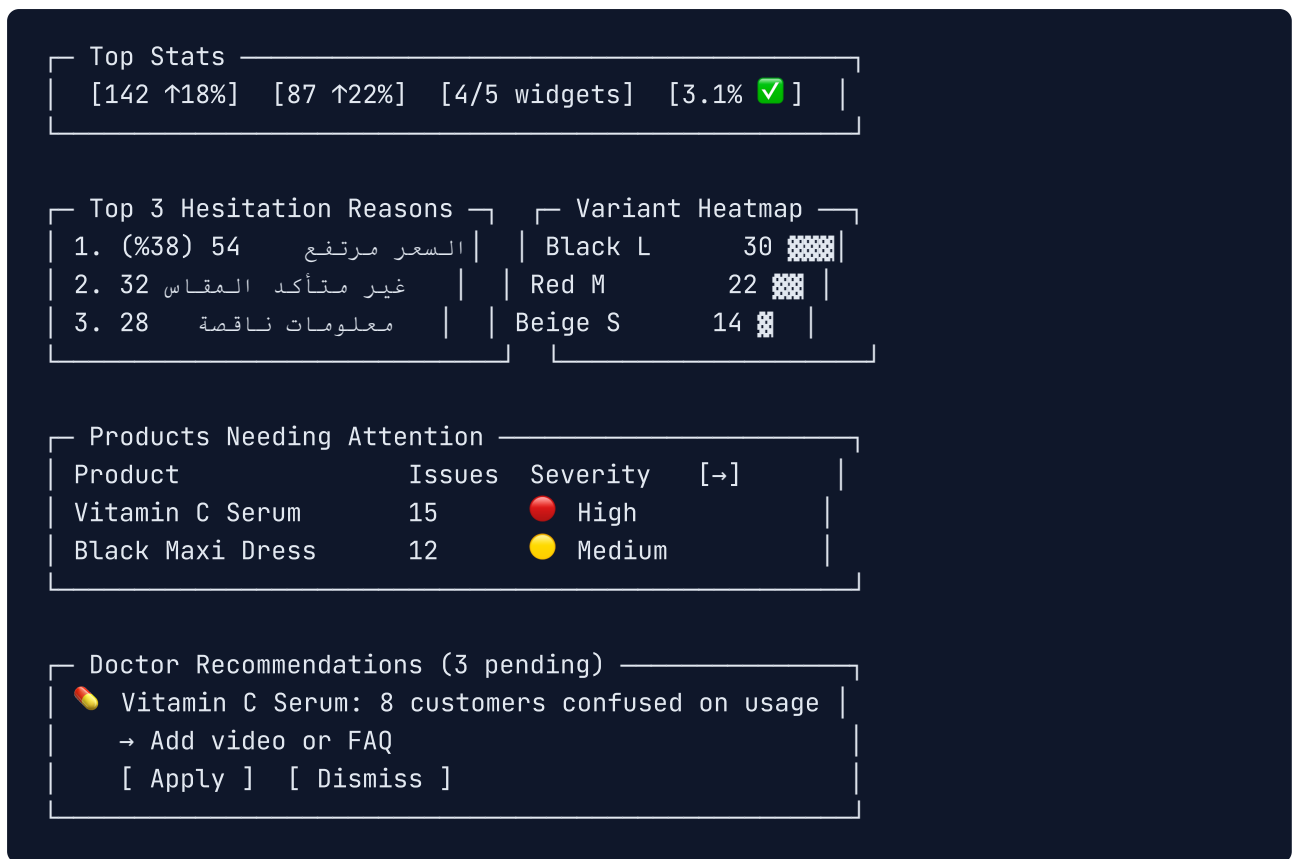
Layout:

Top nav: Logo, store name, plan badge, help, profile menu

Left sidebar: Dashboard, Widgets, Doctor, Analytics, Settings

Main content: card-based layout

Dashboard Home:



Component Library 16.4

Build/use these components:

- `<Button>` (primary, secondary, ghost variants) •
- `<Card>` (with header, body, footer) •
- `<StatTile>` (number + delta + label) •
- `<ProgressBar>` (for submission rate) •
- `<RecommendationCard>` (severity badge + actions) •
- `<EmptyState>` (icon + message + CTA) •
- `<DataTable>` (sortable, paginated) •
- `<SparklineChart>` (7-day trend) •
- `<BarChart>` (top reasons) •
- `<Heatmap>` (variant demand) •
- `<Toast>` (action confirmations) •
- `<Modal>` (configuration dialogs) •

Empty States 16.5

Day 1 (no data):

مرحبًا بك في Smart Snippet System ابدأ بتفعيل أول widget — ستظهر بياناتك خلال 24 ساعة]
[Onboarding ابدأ 🙌

No submissions yet (data <10):

جمعنا 4 ردود حتى الآن نحتاج +10 لإنتاج insights موثوقة [شاركة ال widgets الحالية] 🇸🇦

No recommendations:

كل المنتجات بحالة جيدة هذا الأسبوع سنخبرك حين يحتاج منتج لاهتمامك ✅

Widget rate <1.5%:

معدل تفاعل ضعيف (1.2%) اقتراحات لتحسينه: [3 nudges] 🔴

Onboarding Flow .17

First-Time Setup (5 steps, <5 minutes) 17.1

Step 1: Welcome (auto after install)

- "مرحبًا [Store Name] — لنبدأ بتفعيل أول widget"
- Choose vertical: Beauty / Fashion / Perfumes / Gifts / Other

Step 2: First Widget — Hesitation Capture

- Pre-filled with vertical-specific defaults
- Preview live widget on side
- "هذا ال widget سيظهر للزوار الذين على وشك المغادرة"

- Customize reasons (optional) — or accept defaults
- [Activate Widget]

Step 3: Privacy Policy Setup

- "نوّد لك Privacy Policy template — راجعها وأضفها لمتجرك"
- Generated URL link
- "منشورة Privacy Policy أؤكد أن لدي"

Step 4: Email Preferences

- Pre-filled with Salla account email
- Toggle: weekly digest, recommendations alerts, billing
- "أول digest يصلك صباح الإثنين القادم"

Step 5: Done

- "جاهز! ال widget مفعّل على متجرك"
- Show first thing to expect: "submissions خلال 24 ساعة ستري أول"
- "تريد تفعيل المزيد؟" → [Activate Interest Capture]
- Skip: "Dashboard خذ جولة في"

Empty Dashboard Tour 17.2

After first visit to dashboard with no data:

- Tooltips on each section explaining what'll appear
- "هذا مثال على ما ستري" (Sample data preview (greyed))

Notifications .18

Merchant Notifications 18.1

Weekly Email Digest (Monday 9 AM Saudi):

- Stats summary

- Top 3 hesitation reasons
- 3 Doctor recommendations
- CTA: open dashboard

Real-time Email (opt-in):

- New high-severity Doctor recommendation
- Widget rate dropped below 1.5%
- 10+ customers requesting same variant (urgent reorder signal)

In-App (Dashboard):

- Toast on new recommendation
- Badge on Doctor menu item with count

Customer Notifications 18.2

Triggered (when conditions met):

- WhatsApp message (primary)
- SMS fallback
- Email (if no phone, has email)

Templates (WhatsApp Business):

Template 1 — Price Drop:

مرحبًا 🙌
!المنتج الذي تابعتَه نزل سعره
[Product Name]
السعر السابق: 199 ر.س
🌟 السعر الآن: 149 ر.س
[link]: اشترِ الآن
— [Store Name]
[manage link]: لإلغاء التنبيهات

Template 2 — Variant Available:

مرحبًا 🙌
!اللي طلبته رجع [color/size] الـ

[Product Name]
[Variant details]

اشتر قبل ينفد : [link]

– [Store Name]
لإلغاء التنبيهات : [manage link]

Analytics & Instrumentation .19

Events to Track (Internal Product Analytics) 19.1

Merchant Events:

- app_installed •
- onboarding_step_completed (props: step_number, time_spent) •
- onboarding_completed (props: total_time) •
- widget_created (props: type, vertical_template) •
- widget_activated •
- widget_deactivated •
- dashboard_viewed (props: section) •
- recommendation_applied (props: rule_id, severity) •
- recommendation_dismissed •
- trial_started •
- trial_converted (props: plan, days_to_convert) •
- trial_expired •
- subscription_cancelled (props: reason) •

Customer Events:

- widget_viewed •

widget_interacted	•
widget_submitted	•
widget_dismissed	•
interest_signal_created (props: type)	•
consent_given	•
consent_withdrawn	•

Properties (Common) 19.2

merchant_id	•
merchant_plan	•
store_vertical	•
device_type	•
locale	•
timestamp	•

Internal Dashboards (For Us) 19.3

Acquisition: installs / day, trial starts / day	•
Activation: % of installs that activate first widget within 24h	•
Engagement: weekly active merchants	•
Retention: 7-day, 30-day, 90-day	•
Revenue: MRR, ARR, churn	•
Funnel: install → onboard → first submission → 10 submissions → trial convert	•

Success Metrics .20

North Star Metric 20.1

Weekly Active Merchants with ≥ 1 Applied Recommendation This proves both engagement AND value delivery.

Leading Indicators (Daily/Weekly) 20.2

- New installs per day •
- First-widget activation rate (target $\geq 80\%$ within 24h) •
- Average submission rate per widget (target $\geq 3\%$) •
- Time to first insight (target $< 72\text{h}$ from install) •
- Doctor recommendation apply rate (target $\geq 30\%$) •

Lagging Indicators (Monthly) 20.3

- Trial-to-paid conversion (target $\geq 15\%$) •
- 30-day retention (target $\geq 70\%$) •
- 90-day retention (target $\geq 50\%$) •
- MRR growth (target 25% MoM in first 6 months) •
- NPS (target ≥ 40) •

Kill Criteria (Day 60 from MVP launch) 20.4

- Submission rate $< 1.5\%$ (sustained over 30 days) •
- Trial-to-paid $< 5\%$ •
- 30-day retention $< 40\%$ •
- Weekly dashboard open rate $< 30\%$ •

If any TWO of these hit, pause new feature work and re-evaluate.

Launch Criteria .21

Closed Beta (Day 60) 21.1

Audience: 3-5 invited merchants (Beauty/Perfumes vertical)

Acceptance Criteria:

- All 4 MVP features functional end-to-end •

- Onboarding completes successfully •
 - Hesitation widget submissions captured and visible in dashboard within 1 minute •
 - Doctor generates at least 1 recommendation when threshold met •
 - WhatsApp notifications successfully delivered •
 - PDPL consent layer functional •
 - No P0 bugs •
- Goal:** 1-2 weeks of bug bash + UX feedback

Open Beta (Day 75) 21.2

Audience: 20-30 merchants (invite-only, marketing channels)

Acceptance Criteria:

- Closed beta feedback incorporated •
- Performance budget met (load times) •
- Privacy policy templates finalized in AR + EN •
- All payment flows tested (Salla billing integration) •
- Help docs published •

Salla App Store Submission (Day 90) 21.3

Submission Checklist:

- ☐ App listing in AR + EN •
- ☐ Screenshots (5+ for each: AR, EN) •
- ☐ Demo video (60 seconds) •
- ☐ Privacy Policy URL •
- ☐ Terms of Service URL •
- ☐ Support email + phone •
- ☐ OAuth scopes justified •
- ☐ App pricing configured •
- ☐ Salla App Store review process (typically 2-7 days) •

Public Launch (Day 105 — estimate) 21.4

- Salla App Store approved •
 - Launch announcement (social, email) •
 - 3 case studies published from beta merchants •
 - Customer support ready (response <4h business hours) •
-

Out of Scope (Explicit "No" list for MVP) .22

Features Out of MVP

- ✗ AI-powered recommendations (Phase 3) •
- ✗ Popup builder for general marketing •
- ✗ Email marketing campaigns •
- ✗ SMS marketing campaigns •
- ✗ Helpdesk / ticketing •
- ✗ Loyalty program •
- ✗ Reviews app •
- ✗ Wishlist as standalone •
- ✗ A/B testing engine •
- ✗ Conversion attribution / revenue tracking •
- ✗ Customer segmentation •
- ✗ Audience export to ad platforms •
- ✗ Multi-store accounts •
- ✗ White-label / agency mode •
- ✗ Multi-language beyond AR + EN •
- ✗ Native mobile app (web dashboard only) •
- ✗ Voice / chatbot interface •
- ✗ Custom CSS injection •
- ✗ JavaScript event API for merchants •

- ✗ Direct integrations (Zapier, Make, etc.) •

Technical Out of MVP

- ✗ Multi-region deployment (Saudi Arabia only) •
- ✗ Custom domains for white label •
- ✗ SSO / SAML •
- ✗ Webhooks for merchants (we receive Salla webhooks, but don't send our own to merchants) •
- ✗ Public API for merchants •

Vertical Out of MVP

- ✗ B2B-only stores •
- ✗ Food delivery / restaurants •
- ✗ Service businesses (non-product) •
- ✗ Marketplaces (multi-seller) •

Open Questions .23

These need to be resolved before/during build:

Salla Platform Questions:

- ? Does `product.quantity.low` webhook fire per variant_id or product_id only? **Verify in** .1

Salla Developer Console.

- ? What's the rate limit for Salla Merchant API calls? Plan accordingly for dashboard. .2
- ? Are app snippets sandboxed (Shadow DOM enforced) or full DOM access? .3
- ? Does Salla provide a customer_id on storefront events for logged-in customers? .4
- ? What's the Salla App Store billing flow? Direct merchant billing or Salla-collected? .5
- ? Are there restrictions on widget visual elements (e.g., popups disallowed in some pages)? .6

Product Questions: 7. ? Should we offer "Quick Setup" (defaults) vs "Custom Setup" (full config) for onboarding? 8. ? Should weekly digest be optional from Day 1, or default-on with opt-out? 9. ? Should we show competitor metrics (anonymized benchmarks) in dashboard? Privacy implications? 10. ? Should "Apply" actions go through merchant approval for first 10, then auto-apply later?

Technical Questions: 11. ? Should we use Shadow DOM or iframe for widget isolation? (Shadow DOM = better UX, iframe = better security) 12. ? How do we handle Salla theme updates that might break our injection points? 13. ? What's the failover plan if our backend goes down (don't break merchant's storefront)? 14. ? Should hesitation reasons be merchant-editable from Day 1 or fixed for first 30 days (for cohort analysis)?

Legal / Compliance Questions: 15. ? Do we need a separate Data Processing Agreement with each merchant (PDPL)? 16. ? Can we use merchant data (anonymized) for cross-merchant benchmarks? (probably needs explicit consent) 17. ? What's our SLA on data deletion requests?

Risks & Mitigations .24

R1 — Submission Rate Below Target

Risk: Customers don't engage with widgets (<1.5%) **Severity:** Critical (Fatal Risk) **Probability:** High **Mitigation:**

- A/B test trigger timing in beta •
- Offer optional reward coupon (configurable) •
- Use single-tap options (not free text first) •
- Optimize copy for friendliness, not interrogation •
- Action before build:** Run paper-prototype on 1 merchant for 2 weeks, measure baseline. •

R2 — Salla Native Feature Overlap

Risk: Salla launches similar feature, eroding our value **Severity:** High **Probability:** Medium **Mitigation:**

- Build depth (Doctor rules, variant intelligence) Salla won't replicate quickly •
- Partner with Salla early (pitch as complement) •
- Monitor Salla product announcements weekly •

"R3 — Perceived as "Just Another Popup Tool

Risk: Merchants associate us with PopupSnap, undervaluing **Severity:** High **Probability:** Medium **Mitigation:**

- Strict positioning: "Intent Layer", "Doctor", "Insights" vocabulary •
- Demo emphasizes Dashboard, not widgets •
- Pricing higher than PopupSnap (signals premium) •
- Avoid "popup" in marketing copy •

R4 — PDPL Non-Compliance

Risk: Saudi PDPL fines for improper consent **Severity:** High **Probability:** Low **Mitigation:**

- Legal review before launch •
- Consent layer in Day 1 design •
- Document everything (consent text version, timestamps, IPs) •
- Customer-facing consent management URL •

R5 — Salla Theme Compatibility

Risk: Our widgets break on various themes **Severity:** Medium **Probability:** Medium **Mitigation:**

- Shadow DOM isolation •
- Test on top 10 Salla themes during beta •
- Fallback styles for edge cases •
- Monitor support tickets for theme issues •

R6 — WhatsApp Business API Restrictions

Risk: Meta restricts our notification volume or templates **Severity:** Medium **Probability:** Medium **Mitigation:**

- Pre-approve templates with Meta •
- Use Wati or similar BSP (more flexible than direct Meta) •
- SMS fallback ready •

R7 — Performance Impact on Stores

Risk: Our widget slows down merchant storefronts **Severity:** High **Probability:** Low (with discipline) **Mitigation:**

- Hard performance budget (30KB, 100ms TTI delta) •
- Async loading •
- Self-hosted CDN with edge caching •
- Performance regression tests in CI •

Timeline & Milestones .25

Week-by-Week 90-Day Plan

Phase A: Foundation (Weeks 1-3)

Week 1: Setup & Architecture

- Salla developer account + app registration •
- Repository setup (frontend, backend, dashboard) •
- Database schema implementation •
- CI/CD pipeline •
- Hosting infrastructure (AWS Riyadh) •

Week 2: Salla Integration Core

- OAuth flow with Salla •
- Webhook receivers (price_updated, product_updated, customer_created, app_installed) •
- App snippet registration + injection working on test store •
- Basic Twilight hooks tested •

Week 3: Backend Skeleton

- Public API endpoints (submissions, interest, events) •
- Authentication (JWT for merchant dashboard) •
- Database access layer •
- Rate limiting •
- Error handling + logging (Sentry) •

Phase B: Feature 1 Build (Weeks 4-5)

Week 4: Hesitation Widget — Frontend

- Widget rendering (Shadow DOM) •
- Exit Intent triggers (desktop + mobile fallback) •
- Product Clarity triggers •
- Configuration loading from backend •
- Submission flow •
- Thank you state •
- Mobile responsiveness •
- Arabic + English •

Week 5: Hesitation Widget — Backend + Dashboard

- Submission processing •
- Storage + indexes •
- Basic dashboard view (raw submissions list) •
- Widget configuration UI (CRUD) •
- Per-widget analytics tab •

Phase C: Feature 2 Build (Weeks 6-7)

Week 6: Interest Capture — Frontend

- button on PDP "نبهني" •
- Modal with interest type selection •
- Phone validation (Saudi/Gulf) •

- PDPL consent layer •
- Submission flow •

Week 7: Interest Capture — Backend + Notifications

- Interest signal storage •
- Webhook handlers fire correctly •
- WhatsApp Business integration (BSP setup) •
- SMS fallback (Unifonic) •
- Template approval submitted •
- Customer consent management page •

Phase D: Feature 3 Build (Weeks 8-9)

Week 8: Intent Dashboard

- Dashboard home (stats, top reasons, products) •
- Variant heatmap •
- Date range filtering •
- Empty states •
- Performance optimization (queries) •

Week 9: Product Doctor

- Rules engine (5 MVP rules) •
- Daily cron job (4 AM Saudi) •
- Recommendations CRUD •
- Apply action handlers •
- Recommendations UI •

Phase E: Feature 4 + Polish (Week 10)

Week 10: Analytics + Weekly Digest

- Widget Analytics dashboard •
- Daily aggregation cron •
- Weekly digest email template •

- Email sender setup (SendGrid) •
- First digest test send •

Phase F: Beta Prep (Weeks 11-12)

Week 11: Onboarding + Polish

- 5-step onboarding wizard •
- Empty state tours •
- Help docs (FAQ pages) •
- Privacy Policy template generator •
- Loading states everywhere •

Week 12: Internal QA + Bug Bash

- End-to-end testing on staging •
- Performance testing •
- Cross-browser testing (Chrome, Safari, Firefox) •
- Cross-device testing (iPhone, Galaxy, iPad) •
- Theme compatibility testing (top 5 Salla themes) •
- Security review •
- Closed beta merchants identified + outreach •

Phase G: Beta + Launch (Weeks 13-15)

Week 13: Closed Beta Launch

- 3-5 merchants onboarded with white-glove support •
- Daily check-ins •
- Bug fixes prioritized •

Week 14: Open Beta + Iteration

- Expand to 20-30 merchants •
- Collect feedback •
- Iterate on UX issues •
- Refine Doctor rules based on real data •

Week 15: Salla App Store Submission

- All listing materials prepared •
- Demo video recorded •
- Submission filed •
- Begin reviewing Salla feedback if any •

Critical Path Items

- | | |
|--|----|
| Salla developer account approval (Week 1) — can block start | .1 |
| WhatsApp Business template approval (Week 7-8) — can block notifications | .2 |
| Salla App Store review (Week 15+) — can block launch | .3 |
-

A. Glossary

Definition	Term
Widget asking customer why they didn't buy	Hesitation Capture
Any pre-purchase behavioral data point (hesitation, interest)	Intent Signal
Customer registering for future notification (price, variant)	Interest Signal
Rule-based engine recommending fixes per product	Product Doctor
A customer's response to a hesitation widget	Submission
Submissions ÷ Widget Views	Submission Rate
Saudi Personal Data Protection Law (effective 2023-2024)	PDPL
Salla's theme engine + JS SDK	Salla Twilight
Salla's mechanism for injecting JS/HTML into storefronts	App Snippet
Salla feature to filter webhooks by rules	Conditional Webhook

B. Sample Doctor Rule Configuration (Future Extensibility)

```
// Rules will be database-driven for easy iteration
interface DoctorRuleDb {
  id: string;
  name: string;
  enabled: boolean;
  conditions: ConditionExpression; // composable AND/OR
  recommendation_template: {
    title_template_ar: string; // with {count}, {product_name} placeholders
    title_template_en: string;
    explanation_ar: string;
    explanation_en: string;
    action_type: string;
    action_payload_template: any;
  };
  severity: 'low' | 'medium' | 'high';
  cooldown_days: number; // don't re-recommend within X days
  min_signal_count: number; // threshold
  vertical_filter?: string[]; // only for specific verticals
}
```

C. Hesitation Reason Taxonomy (Default)

```
const DEFAULT_REASONS = [
  { id: 'price_high', label_ar: 'السعر مرتفع', label_en: 'Price too high', cat
  { id: 'shipping_unclear', label_ar: 'الشحن غير واضح', label_en: 'Shipping un
  { id: 'size_unclear', label_ar: 'غير متأكد من المقاس', label_en: 'Unsure abo
  { id: 'comparing', label_ar: 'محتار بين منتجات', label_en: 'Comparing produc
  { id: 'need_more_info', label_ar: 'محتاج معلومات أكثر', label_en: 'Need more
  { id: 'just_browsing', label_ar: 'فقط أتصفح', label_en: 'Just browsing', cat
  // Vertical-specific (added in v2):
  { id: 'ingredients_concern', label_ar: 'قلق من المكونات', label_en: 'Concern
  { id: 'authenticity', label_ar: 'متشكك في الأصالة', label_en: 'Authenticity c
];
```

D. Sample API Payloads (Detailed)

See section 15 for complete spec. Additional examples:

Webhook payload from Salla (`product.price.updated`):

```
{
  "event": "product.price.updated",
  "merchant": 12345,
  "created_at": "2026-05-12T14:30:00Z",
  "data": {
    "id": 67890,
    "name": "Vitamin C Serum",
    "old_price": 199.00,
    "new_price": 149.00,
    "currency": "SAR"
  }
}
```

Doctor Recommendation creation:

```
{
  "rule_id": "rule_size_confusion",
  "product_id": "67890",
  "severity": "high",
  "title_ar": "8 عملاء Vitamin C Serum غير متأكدين من المقاس على 8",
  "title_en": "8 customers unsure about size on Vitamin C Serum",
  "explanation_ar": "أو دليل المقاسات Size Help snippet أضف",
  "explanation_en": "Add a Size Help snippet or size chart",
  "action_type": "create_snippet",
  "action_payload": {
    "snippet_template": "size_help"
  },
  "signal_count": 8,
  "status": "pending",
  "expires_at": "2026-05-26T14:30:00Z"
}
```

E. References

- Strategic Analysis:** [Smart-Snippet-System-Analysis.md](#) •
- Competitive Landscape:** [Competitive-Landscape-Analysis.md](#) •
- Salla Developer Docs:** <https://docs.salla.dev/> •
- Salla Webhooks:** <https://docs.salla.dev/421119m0> •
- Twilight Theme Hooks:** <https://docs.salla.dev/422552m0> •
- PDPL Compliance:** <https://sdaia.gov.sa/en/Research/Pages/DataProtection.aspx> •

Module 40 + Phase 1/2 Roadmap Specifications .27

This section adds explicit specs for 20 modules referenced in the Analysis library but not covered in MVP Features 1-4:

- **Module 40** (PDPL Consent Center) — formally MVP-scope requirement
- **8 Phase 1 modules** (Months 4-9)
- **11 Phase 2 modules** (Months 10-18)

Specs are condensed (Description + Top FRs + Salla Integration + Dependencies + Effort). Full UX/edge case treatment will be added when each module enters active development.

Module 40 — PDPL Consent Management Center (🟢 MVP — Legal 27.1 Requirement)

Description: Customer-facing self-service portal for managing data, signals, and consent.
Required for PDPL compliance (already referenced as cross-cutting requirement in Section 11.2; this section formalizes it as a specific feature).

Functional Requirements:

- FR-40.1: Accessible via `/v1/public/consent/manage?token=<jwt>`. Token has 30-day expiry, regenerable.
- FR-40.2: Display all interest signals + submissions associated with customer's phone hash.
- FR-40.3: One-click cancel for individual signals (sets `status='cancelled'`).
- FR-40.4: Bulk delete with confirmation prompt (cascade delete + audit log).
- FR-40.5: Consent withdrawal (stops future communications, retains audit record).
- FR-40.6: Data export (CSV/JSON download).
- FR-40.7: Log all actions to `consent_records` table with timestamp, IP, action type.
- FR-40.8: Email confirmation after each major action (delete, withdraw).
- FR-40.9: Arabic + English (RTL support).

Salla Integration: None (self-hosted URL). **Dependencies:** Database schema

(`interest_signals` , `consent_records` tables) — already in MVP. **Effort:** Medium (3-5 days).

Open Questions: Should we require phone OTP for full data deletion to prevent token-leak attacks?

Phase 1 Modules (Months 4-9) 27.2

All Phase 1 modules build directly on MVP data infrastructure. Technically verified, low Salla native overlap.

Module 38 — Empty Search Capture (🟡 Phase 1)

Description: Captures customer search queries that returned no results. Direct inventory demand signal. **Key FRs:**

- FR-38.1: Detect empty search state via `.s-search-no-results` CSS class or DOM observation. •
- FR-38.2: Inject capture form: pre-filled query + optional phone + PDPL consent. •
- FR-38.3: Store in `missed_searches` table with frequency aggregation. •
- FR-38.4: Dashboard widget: "Top missed searches" sorted by count. •

Salla Integration: Twilight `<salla-search>` component customization (*Confirmed*).

Dependencies: PDPL consent layer (MVP). **Effort:** Small-Medium (2-4 days).

Module 39 — First-Time vs Returning Recognition (🟡 Phase 1)

Description: Different greeting/widget for new vs returning visitors, with intent pre-tagging.

Key FRs:

- FR-39.1: Detect visitor state via cookie + Salla customer login state. •
- FR-39.2: First-time: show 3-option intent selector (gift / personal / browse). •
- FR-39.3: Returning: show last-viewed product + recently viewed strip. •
- FR-39.4: Store intent tag in `visitor_intents` table linked to session. •

Salla Integration: Twilight SDK for customer state (*Confirmed*). **Dependencies:** None (independent feature). **Effort:** Medium (4-5 days).

Module 46 — Vertical Discovery Quiz (🟡 Phase 1)

Description: Short quiz building customer preference profile (skin type, style, scent family).

Key FRs:

- FR-46.1: Configurable quiz templates per vertical (beauty, fashion, perfumes). •
- FR-46.2: 3-5 question flow, < 60 seconds completion. •
- FR-46.3: Generate `customer_profile` JSON stored in `customer_profiles` table. •
- FR-46.4: Apply profile to product filtering + recommendations. •

Salla Integration: Products API for filtering. **Dependencies:** Module 39 (First-Time recognition) for trigger. **Effort:** Medium-Large (5-7 days).

Module 48 — Recently Viewed Memory Strip (🟡 Phase 1 — overlap care)

Description: Sticky horizontal strip showing recently viewed products + Dashboard tracking.

Key FRs:

- FR-48.1: Store last 20 viewed products in localStorage per visitor. •
- FR-48.2: Display sticky strip on PDP/Cart with horizontal scroll. •
- FR-48.3: Track view-to-purchase conversion per product in `widget_events`. •

Salla Integration: Twilight body hooks (*Confirmed*). **Dependencies:** Module 35 (Analytics infrastructure). **Effort:** Small-Medium (3-4 days). **Note:** Some Salla themes already show "recently viewed" natively. Position our snippet to add tracking value, not duplicate visual.

Module 49 — Birthday/Anniversary Capture (🟡 Phase 1)

Description: Capture personal dates for triggered campaigns. **Key FRs:**

- FR-49.1: Form fields: birthday (DD/MM), anniversary (optional, DD/MM). •
- FR-49.2: PDPL-compliant consent for date storage. •
- FR-49.3: Daily cron checks dates, triggers WhatsApp/Email 3 days before. •
- FR-49.4: Dashboard: "Birthdays this month" with campaign launcher. •

Salla Integration: Customer fields (*verify if natively supported*). **Dependencies:** Module 7 (Interest Capture form components). **Effort:** Small (2-3 days). **Verification Needed:** Whether Salla `customer` object has birthday/anniversary fields natively.

Module 51 — Coming Soon / Pre-Launch Capture (🟡 Phase 1)

Description: Capture pre-launch interest for unpublished products. **Key FRs:**

- FR-51.1: Detect product status="hidden" or custom "coming_soon" tag. •
- FR-51.2: Replace standard PDP with pre-launch teaser + email/phone capture. •
- FR-51.3: Optional "Early Bird discount" coupon assignment. •
- FR-51.4: Auto-notify all subscribers when product status changes to "active". •

Salla Integration: Products API + status webhooks (*Confirmed*). **Dependencies:** Module 8 (Notification infrastructure). **Effort:** Medium (4-5 days).

Module 54 — Influencer Code Capture & Attribution (🟡 Phase 1)

Description: Capture influencer-specific traffic, track per-influencer revenue. **Key FRs:**

- FR-54.1: Parse UTM params + special URL slugs (`/?ref=sara_kn`). •
- FR-54.2: Display personalized landing message with influencer name. •
- FR-54.3: Auto-apply influencer coupon code at checkout. •
- FR-54.4: Dashboard: per-influencer revenue, AOV, conversion rate, CAC. •

Salla Integration: Coupons API for auto-apply (*Confirmed*). **Dependencies:** None. **Effort:** Medium (4-6 days).

Module 55 — Cart Sharing & Save Link (🟡 Phase 1)

Description: Generate shareable cart URL for WhatsApp/Email sharing with dual reward. **Key FRs:**

- FR-55.1: Generate signed JWT containing cart state. •
- FR-55.2: Public URL `/cart/share/<token>` reconstructs cart on visit. •
- FR-55.3: Track shared cart conversions with referral attribution. •
- FR-55.4: Optional 10% discount for both sharer and buyer upon purchase. •

Salla Integration: Cart API for state reconstruction. **Dependencies:** Module 22 (Cart-related infrastructure from Phase 1). **Effort:** Medium (4-5 days).

Phase 2 Modules (Months 10-18) 27.3

Phase 2 modules require deeper data, technical verification, or careful positioning.

Module 37 — Checkout Hesitation Capture (🟡 Phase 2)

Description: Capture abandonment reasons specifically on checkout page. **Key FRs:**

- FR-37.1: Detect exit intent on checkout page (mouseleave / scroll-up / idle). •
- FR-37.2: Show 6-option reason selector (shipping, payment, security, etc.). •
- FR-37.3: Aggregate in `checkout_hesitations` table separate from PDP hesitations. •
- FR-37.4: Trigger different Doctor rules vs PDP hesitations. •

Salla Integration: Twilight checkout hooks (⚠️ **NEEDS VERIFICATION**). **Dependencies:** Feature 1 (Hesitation Capture) for shared components. **Effort:** Medium-Large (5-7 days, contingent on Salla checkout customization).

Module 41 — Live Stock Urgency Signal (🟡 Phase 2 — overlap care)

Description: Transparent stock count + social proof ("X left + Y customers viewing"). **Key FRs:**

- FR-41.1: Subscribe to `product.quantity.low` webhook (*Confirmed*). •
- FR-41.2: Display badge only when stock $\leq$ configurable threshold (default 5). •
- FR-41.3: Augment with real-time viewing count from analytics events. •
- FR-41.4: Skip display if Salla theme already renders stock count (theme detection). •

Salla Integration: Webhook + Twilight PDP hooks (*Confirmed*). **Dependencies:** Webhook handlers from MVP. **Effort:** Small-Medium (3-5 days).

Module 42 — Mobile One-Tap Interest (🟡 Phase 2 — overlap with Salla Wishlist)

Description: Anonymous mobile-only interest capture without form. Companion to (not replacement of) Salla Wishlist. **Key FRs:**

- FR-42.1: Display sticky heart icon on PDP mobile only. •
- FR-42.2: Save product ID to cookie + send anonymous event. •
- FR-42.3: On return visit, restore from cookie with reminder banner. •
- FR-42.4: Aggregate "most anonymous-saved products" for Doctor. •

Salla Integration: Twilight SDK + mobile detection. **Dependencies:** Analytics infrastructure (MVP). **Effort:** Small (2-3 days). **Kill Criterion:** If incremental volume vs Salla native Wishlist < 2x in pilot, remove from roadmap.

Module 43 — Pre-Checkout Confidence Booster (🟡 Phase 2)

Description: Trust signals + WhatsApp CTA on checkout page above "Place Order". **Key FRs:**

- FR-43.1: Inject configurable trust block (shipping, returns, payment security). •
- FR-43.2: Optional last-minute WhatsApp support button. •
- FR-43.3: Track impression/click-to-WhatsApp conversion. •

Salla Integration: Twilight checkout hooks (⚠️ **NEEDS VERIFICATION**). **Dependencies:** Same checkout verification as Module 37. **Effort:** Small (2-3 days, contingent on hooks).

Module 44 — Smart Reorder Timing (🟡 Phase 2)

Description: Auto-triggered reorder reminders based on product category timing. **Key FRs:**

- FR-44.1: Define `reorder_category_timings` config table (e.g., serum=60d, perfume=90d). •
- FR-44.2: Daily cron scans completed orders, identifies due reorders. •
- FR-44.3: Send WhatsApp reminder with one-click reorder link. •
- FR-44.4: Track reorder rate per category for tuning. •

Salla Integration: Orders API + WhatsApp BSP. **Dependencies:** WhatsApp infrastructure from MVP. **Effort:** Medium (5-7 days).

Module 45 — Comparison Shopping Detector (🟡 Phase 2)

Description: Detect customers viewing 3+ similar products, intervene with help. **Key FRs:**

- FR-45.1: Track PDP visits per session in real-time (server-side). •
- FR-45.2: When session shows 3+ PDPs from same category within 10 minutes, trigger widget. •
- FR-45.3: Widget asks: "What's making the decision hard?" with 5 options. •
- FR-45.4: Surface insights in Doctor for category-level optimization. •

Salla Integration: Products API for category matching. **Dependencies:** Session tracking infrastructure (Module 35 Analytics). **Effort:** Medium-Large (6-8 days).

Module 47 — Customer Bundle Builder (🟡 Phase 2)

Description: Customer selects 3+ products to form custom bundle with auto-discount. **Key FRs:**

- FR-47.1: Configurable rules: min/max products, eligible categories, discount tiers. •
- FR-47.2: Bundle UI with selected items, dynamic pricing, "complete bundle" CTA. •
- FR-47.3: Apply discount as Salla coupon at checkout. •
- FR-47.4: Dashboard: "Top customer-built combinations" to inform official bundles. •

Salla Integration: Coupons API + Products API. **Dependencies:** None. **Effort:** Large (8-10 days).

Module 50 — Out-of-Stock Substitute Engine (🟡 Phase 2)

Description: Suggest substitutes immediately when product is OOS + capture preference.

Key FRs:

- FR-50.1: Detect OOS state via stock = 0 check. •
- FR-50.2: Query Products API for similar products (same category, price, attributes). •
- FR-50.3: Display 3 alternatives + "wait for original" option. •
- FR-50.4: Track substitute conversion rate vs wait rate. •

Salla Integration: Products API for similarity queries. **Dependencies:** None. **Effort:** Medium (5-6 days).

Module 52 — Smart Coupon Discovery (🟡 Phase 2)

Description: Customer-facing coupon search + capture of searched non-existent codes. **Key FRs:**

- FR-52.1: Search bar in cart/checkout: "Looking for a coupon?" •
- FR-52.2: Suggest available eligible coupons based on cart contents. •
- FR-52.3: Capture all searched code names (even non-existent) in `coupon_searches` table. •
- FR-52.4: Dashboard: "Searched codes not in system" → uncovers ghost influencer codes. •

Salla Integration: Coupons API. **Dependencies:** None. **Effort:** Medium (4-6 days).

Module 53 — Shipping Transparency Calculator (🟡 Phase 2)

Description: Pre-checkout shipping cost calculator on PDP/Cart. **Key FRs:**

- FR-53.1: City/region dropdown with auto-detect from IP. •
- FR-53.2: Display shipping options + cost in real-time. •
- FR-53.3: "Add X SAR for free shipping" progress bar. •
- FR-53.4: Track shipping objection rate vs original cart abandonment. •

Salla Integration: Shipping API (⚠️ **VERIFY AVAILABILITY**). **Dependencies:** None. **Effort:** Medium (5-7 days, contingent on API).

Module 56 — Free Sample / Trial Request (🟡 Phase 2)

Description: Sample request workflow for high-AOV products (>500 SAR). **Key FRs:**

- FR-56.1: Eligible product detection (price threshold + sample-eligible tag). •
- FR-56.2: Sample request form with address + phone. •
- FR-56.3: Create Salla order with sample SKU (custom flow). •
- FR-56.4: Track sample-to-purchase conversion + revenue attribution. •

Salla Integration: Orders API. **Dependencies:** Order management infrastructure. **Effort:** Large (8-10 days, includes operational workflow).

Roadmap Total Effort Estimate 27.4

Total Effort	Modules	Phase
3-5 days	1	MVP (Module 40 explicit)
25-35 days	8	Phase 1 (38, 39, 46, 48, 49, 51, 54, 55)
55-75 days	11	Phase 2 (37, 41, 42, 43, 44, 45, 47, 50, 52, 53, 56)
80-110 dev days (16-22 weeks)	20	Total

Critical Path Items 27.5

These can block multiple modules:

1. **Salla checkout customization verification** — blocks 37, 43 (potentially 53).
2. **WhatsApp Business API setup** — blocks 44, 49, 51 notifications.
3. **Salla customer fields availability** — blocks 49 (Birthday).
4. **Session tracking infrastructure** — blocks 45 (Comparison Detector).

Out of Scope Notes for Roadmap 27.6

- Modules listed are **NOT** part of MVP scope. Building prematurely risks MVP timeline.
- Each module must pass MVP kill criteria before development begins.
- Re-prioritize roadmap quarterly based on real merchant feedback.

Document Sign-Off

Status	Date	Name	Role
Pending	TBD	[Founder]	Product Owner
Pending	TBD	TBD	Engineering Lead
Pending	TBD	TBD	Design Lead
Pending	TBD	TBD	Legal Review

PRD — MVP نهاية

التحدي الأكبر: الانضباط بـ MVP. أي feature إضافي = 1- من الأربعة الأساسية. **Success:** ال Submission rate $\geq 3\%$ + Trial-to-paid $\geq 15\%$ + 30-day retention $\geq 70\%$ **Timeline:** 15 من Day 1 إلى Salla App Store submission.